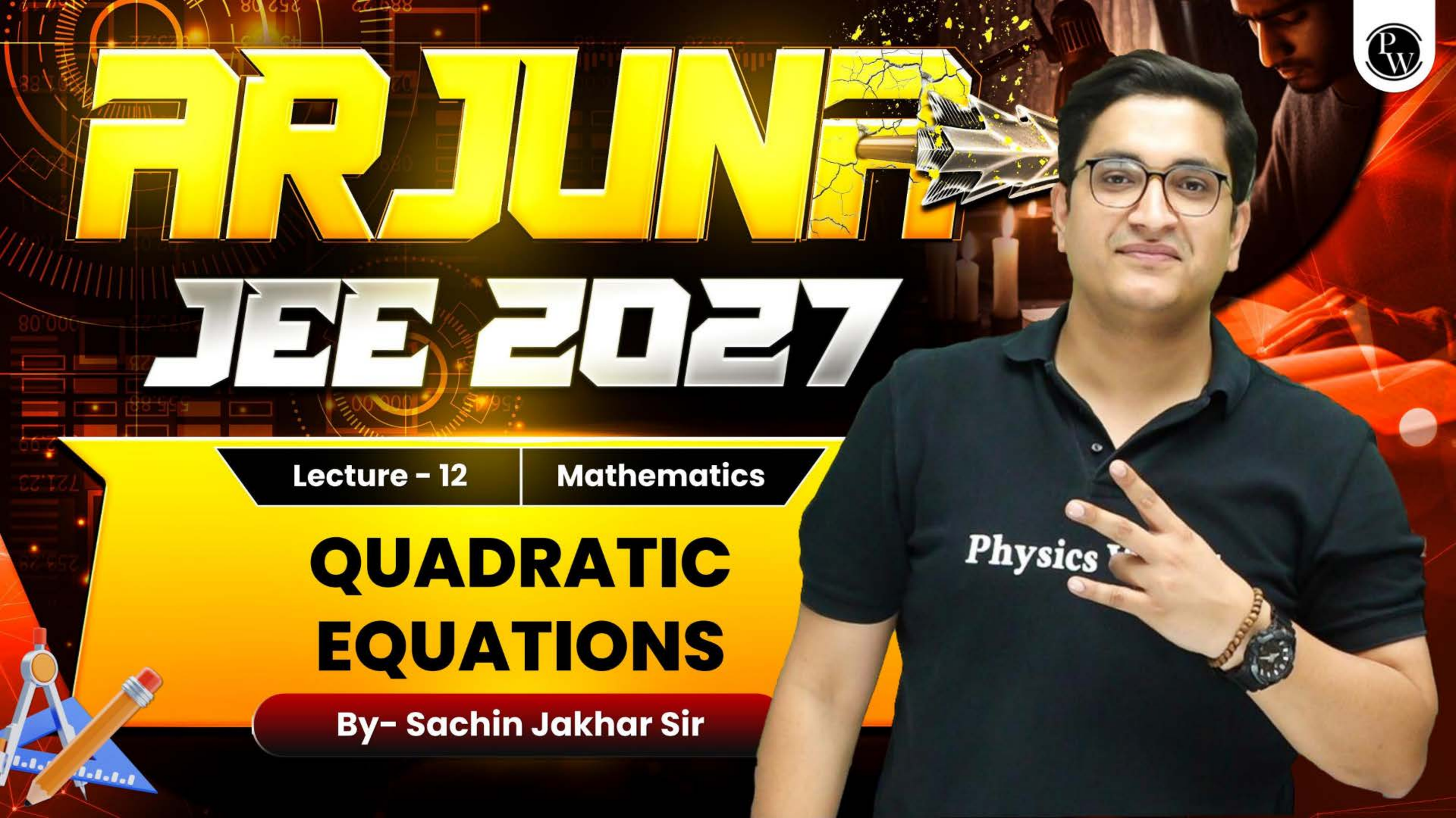
ARJUNA JEE 2027

Lecture - 12 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Location of Roots**

B **Theory of Higher Degree Equations**

C **Question Practice**

D





LAST CLASS REVISION

Answer the following Questions:

1. If $(x = \alpha)$ is common root of two polynomial equations $P_1(x) = 0$ and $P_2(x) = 0$ then:

   (a) $(x = \alpha)$ will also be the root of $mP_1(x) \pm nP_2(x) = 0$. → (T)

   (b) All the roots of $mP_1(x) \pm nP_2(x) = 0$, are common roots. → (F)

2. If exactly one root is common between two quadratics, then what is the method to proceed?

   • Apply cond^n of atleast 1 common
   → Check!

Question Practice

Acidic Question

If the quadratic equations $x^2 + ax + 12 = 0$ and $x^2 + bx + 15 = 0$ and $x^2 + (a+b)x + 36 = 0$ have a common positive root find 'a' and 'b' and the root of the equation.

$x^2 + ax + 12 = 0 \rightarrow \alpha$

$x^2 + bx + 15 = 0 \rightarrow \alpha$

add

$2x^2 + (a+b)x + 27 = 0 \rightarrow \alpha$

$x^2 + (a+b)x + 36 = 0 \rightarrow \alpha$

$\ominus$

$x^2 - 9 = 0 \rightarrow \alpha$

$x = \pm 3$

$x = 3$ is C.R.

$x = 3$

$9 + 3a + 12 = 0$

$a = -\frac{21}{3} = -7$

$x = 3$

$9 + 3b + 15 = 0$

$b = -\frac{24}{3} = -8$

Question [AIEEE-2008]

71%

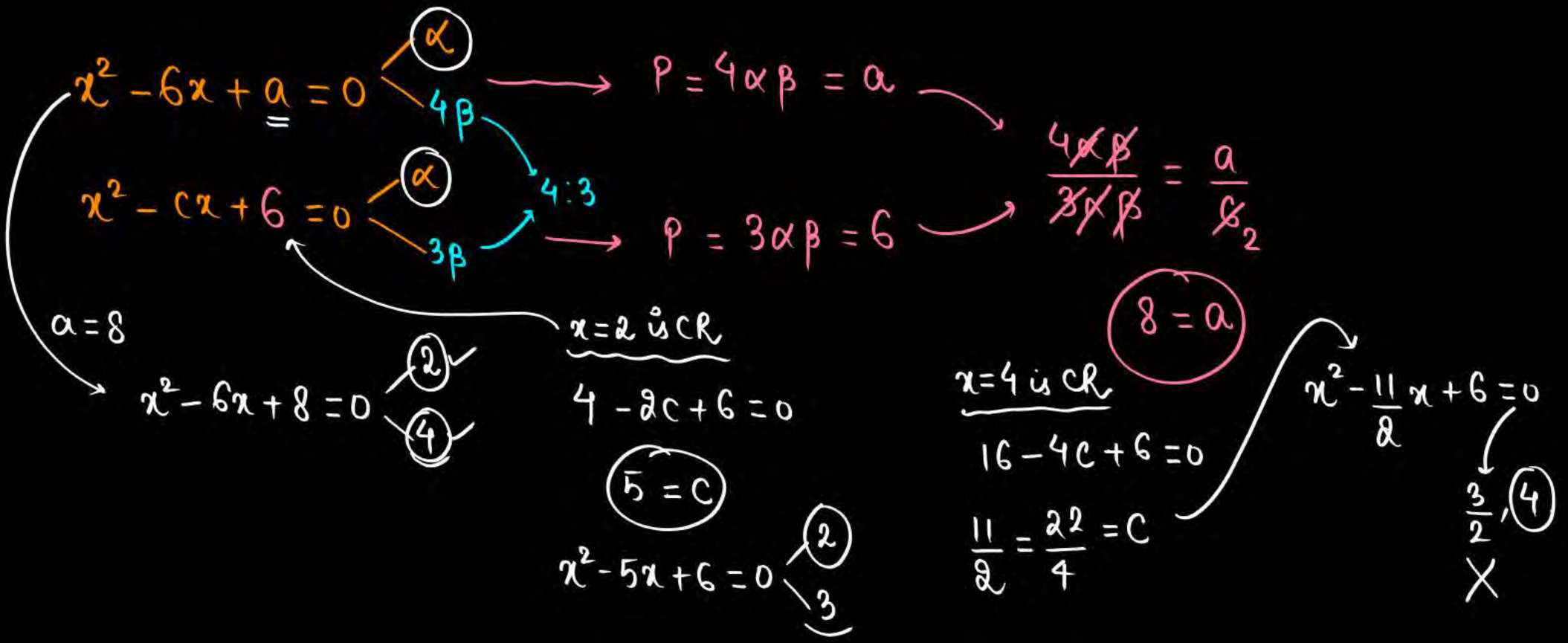
The quadratic equations $x^2 - 6x + a = 0$ and $x^2 - cx + 6 = 0$ have one root in common. The other roots of the first and second equations are integers in the ratio $4 : 3$. Then the common root is

A 1

B 4

C 3

D 2



$b = 0, 2, -2$

Question

Let $f(x) = x^3 - 3x + b$ and $g(x) = x^2 + bx - 3$, where b is a real number. If the equation $f(x) = 0$ and $g(x) = 0$ have a common root, then

A. Number of possible values of b is 3.

B. Number of possible values of b is 2.

C. Sum of all possible values of b is 0.

D. Sum of all possible values of b is 2.

$x = -1$ is CR

$b = -2$ ✓

$x^3 - 3x - 2 = 0$

$x^2 - 2x - 3 = 0$

$\times x$

$x^3 - 3x + b = 0$

$x^3 + bx^2 - 3x = 0$

$-bx^2 - 3x + 3x + b = 0$

$b(1 - x^2) = 0$

$b = 0$ OR $x = \pm 1$

$x^3 - 3x = 0 \Rightarrow 0, \sqrt{3}, -\sqrt{3}$

$x^2 - 3 = 0 \Rightarrow \sqrt{3}, -\sqrt{3}$

$x = 1$ is CR

$b = 2$ ✓

$x^3 - 3x + 2 = 0 \rightsquigarrow x = 1$

$x^2 + 2x - 3 = 0 \rightsquigarrow x = 1$

Acidic Question

If the equations $x^3 + x^2 - 4x = 4$ and $x^2 + px + 2p = 0\ (p \in R)$ have two roots common, then the value of p is



A -2

B -1

C 1

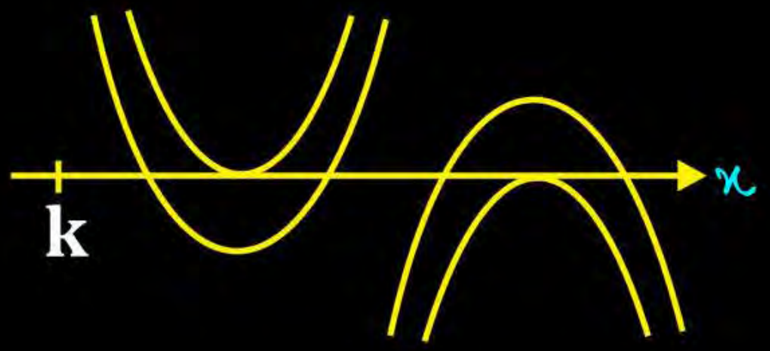
D 3

LOCATION OF ROOTS

RAR

Case-I

If both roots are greater than number given number 'k'



1. $D \geq 0$
2. $\frac{-b}{2a} > k$
3. $af(k) > 0$

$\cap$

Case-II

If both roots are less than number given number 'k'

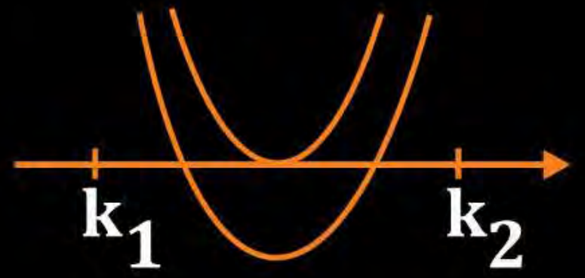


1. $D \geq 0$
2. $\frac{-b}{2a} < k$
3. $af(k) > 0$

$\cap$

Case-III

Both roots lies between 'k_1' & 'k_2'.



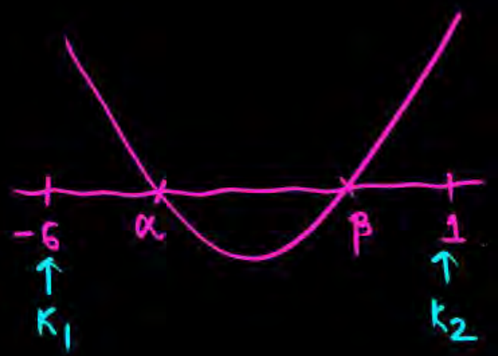
1. $D \geq 0$
2. $k_1 < \frac{-b}{2a} < k_2$
3. $af(k_1) > 0$
4. $af(k_2) > 0$

$\cap$

Question #Lengthy !!

Ans.: $k \in \left(6, \frac{27}{4}\right)$

If α, β are the roots of the quadratic equation $\underbrace{x^2 + 2(k-3)x + 9}_{f(x)} = 0$ $(\alpha \neq \beta)$.
If $\alpha, \beta \in (-6, 1)$, then find the values of k.

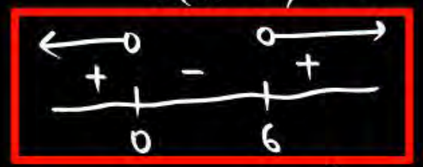


① $D > 0$

$4((k-3))^2 - 4 \times 9 > 0$

$k^2 - 6k + 9 - 9 > 0$

$k(k-6) > 0$



② $-6 < \frac{-b}{2a} < 1$

$-6 < \frac{-(2)(k-3)}{2 \times 1} < 1$

$6 > k-3 > -1$

$9 > k > 2$

③ $af(-6) > 0$

$1.(36 + 2(k-3)(-6) + 9) > 0$

$3(12 - 4(k-3) + 3) > 0$

$12 - 4k + 12 + 3 > 0$

$27 > 4k$

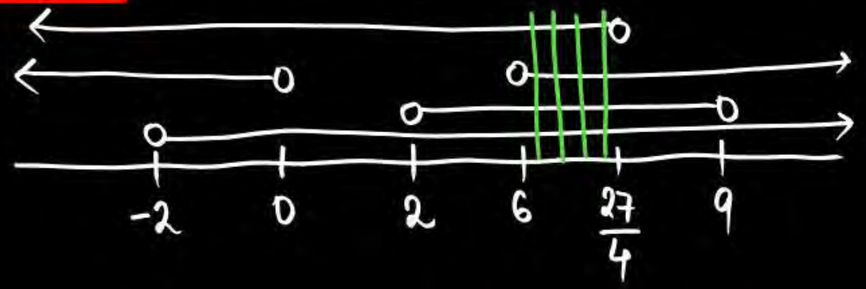
$\frac{27}{4} > k$

④ $af(1) > 0$

④ $1.(1 + 2(k-3) + 9) > 0$

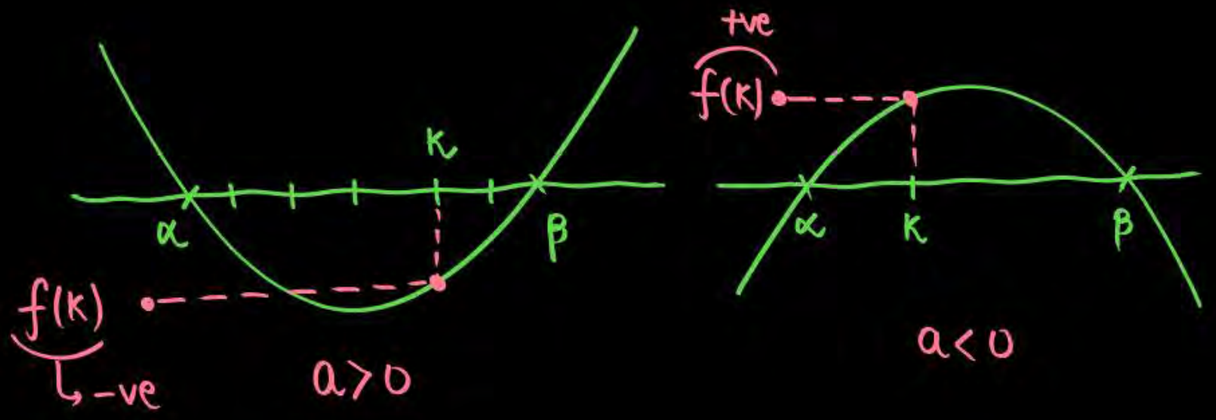
$2k + 4 > 0$

$k > -2$



LOCATION OF ROOTS

Case-IV : Any number k lies between the roots:





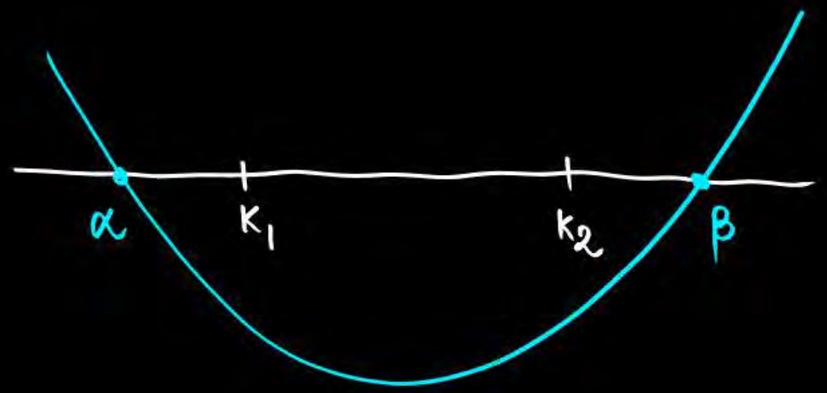
Note :-

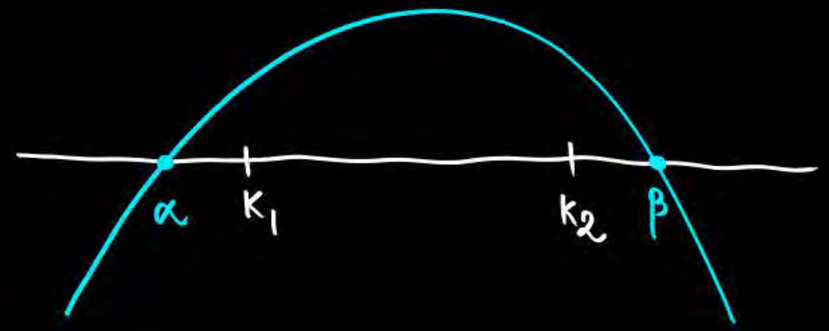
If one root is less than k_1

& other root is greater than k_2.

$af(k_1) < 0$

$af(k_2) < 0$



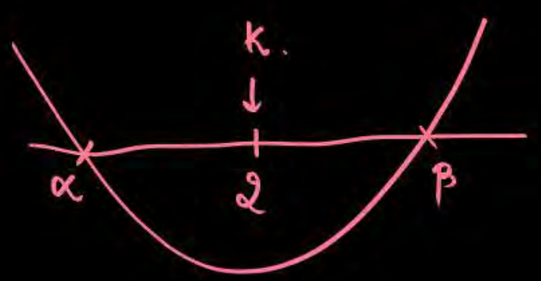


Question

Ans. : $k \in (-2, 3)$

Find the value of k for which one root of the equation of $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceed 2 and other is smaller than 2 .

$f(x)$



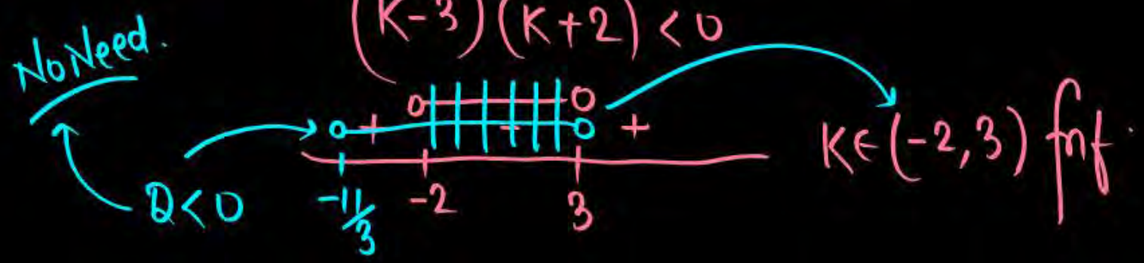
$af(2) < 0$

$$1.\ \{4 - (k+1)2 + k^2 + k - 8\} < 0$$

$$(4 - 2k - 2 + k^2 + k - 8) < 0$$

$$k^2 - k - 6 < 0$$

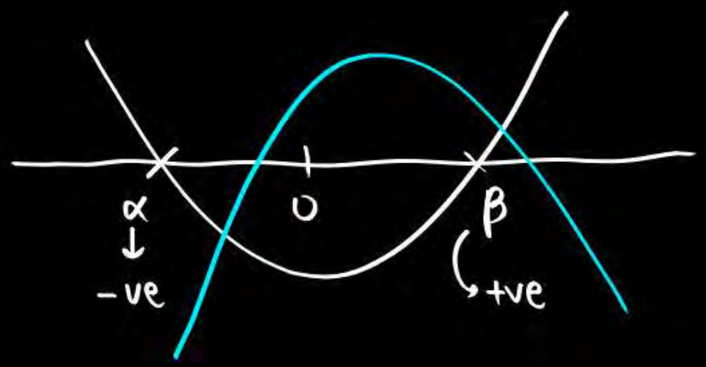
$$(k-3)(k+2) < 0$$



Language: $y = ax^2 + bx + c$

↳ Roots are of opp. sign

↳ zero lies b/w Roots.



$a\,f(0) < 0$

$a \cdot c < 0$

ex: $1x^2 - 5x - 6 = 0$ → +ve, -ve

ex: $2x^2 + 11x - 3 = 0$ → +ve, -ve

Question

Ans. : $k \in \left(\frac{8}{3}, 5\right)$

Find the value of k for which one root of the equation of $(k-5)x^2 + 2kx + k - 4 = 0$ is smaller than 1 and the other root exceed 2.

$f(x)$.



$(K-5) \cdot f(1) < 0$ — (1)

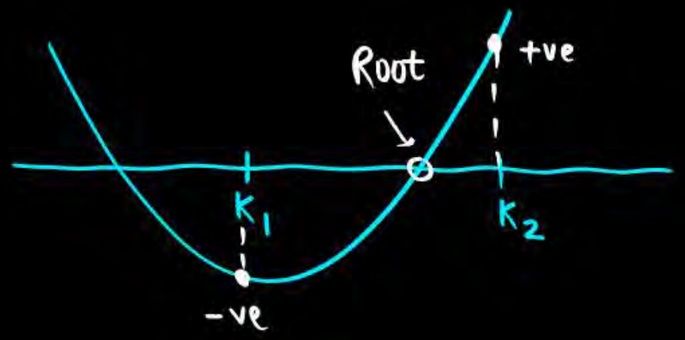
$(K-5) \cdot f(2) < 0$ — (2)

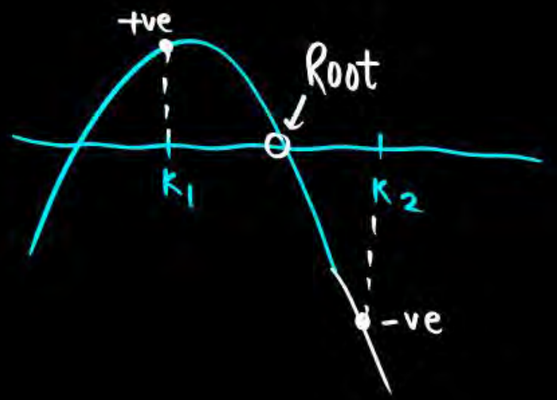
→ $\cap$

DIBY!!

LOCATION OF ROOTS

Case-V : If exactly one root of quadratic lies between k_1 & k_2 :



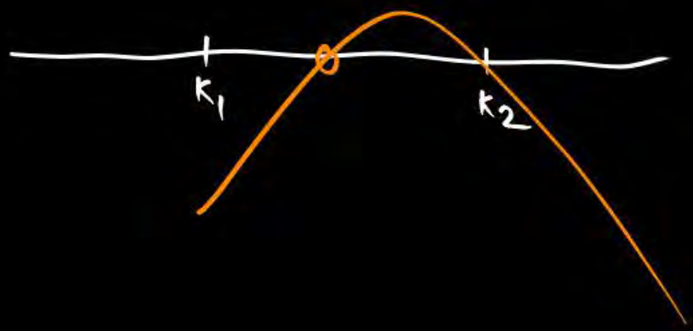


$f(k_1) \cdot f(k_2) < 0$

N & S.

Boundary condⁿ.





$x = k_1$ is a root.

↓

Quad. $\vec{H}$ satisfy

↓

get parameter

↓

Calculate Roots.

IMPORTANT NOTE

In Case-V, Boundary Condition should be separately analyzed.

Boundary Condition:

If $x = k_1$ or $x = k_2$ is a root of Quadratic.

Step-01: Put $x = k_1$ or $x = k_2$ in given quadratic and find parameter.

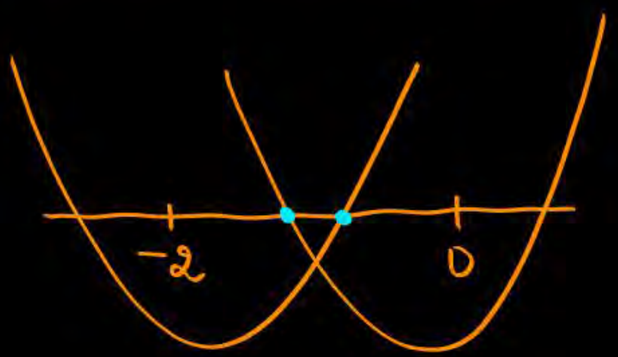
Step-02: Substitute value of parameter in given Quadratic and calculate both roots.

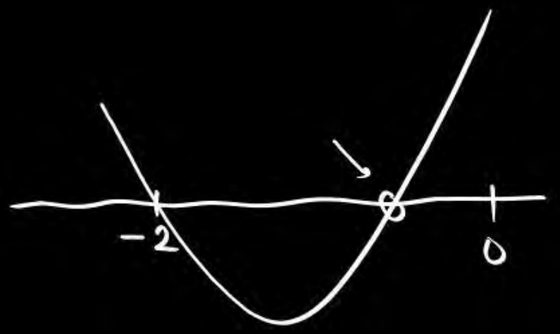
Step-03: Now check, does roots satisfy given condition. If yes, value of parameter is included. If no, value of parameter is excluded.

Question

Find all possible values of m for which exactly one root of the equation $x^2 + mx + m^2 + 6m = 0$, lies in $(-2, 0)$.

k_1 k_2





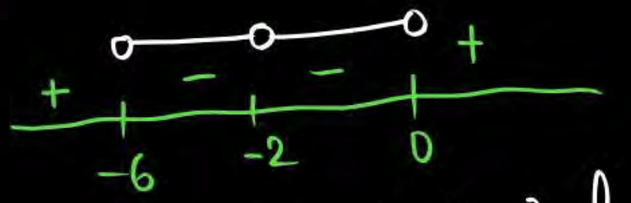
$f(-2) \cdot f(0) < 0$

$x=-2$ $x=0$

$$(4 - 2m + m^2 + 6m)(m^2 + 6m) < 0$$

$$(m^2 + 4m + 4)\, m(m+6) < 0$$

$$m(m+6)(m+2)^2 < 0$$



$m \in (-6, 0) - \{-2\}$ Ans.

Question

Find all possible values of m for which exactly one root of the equation $x^2 + mx + m^2 + 6m = 0$, lies in $(-2, 0)$.

Bound. cond^n



$x = -2$ is a root. ✓

$\Rightarrow 4 - 2m + m^2 + 6m = 0$

$(m+2)^2 = 0$ ✓

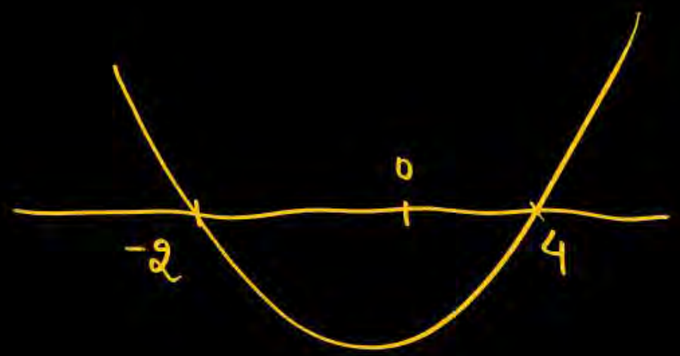
$\# \quad m = -2$ ✓

Rejected.

Quad.

$x^2 - 2x - 8 = 0$

$(x-4)(x+2) = 0 \quad \begin{matrix} 4 \\ -2 \end{matrix}$



Question

Find all possible values of m for which exactly one root of the equation $x^2 + mx + m^2 + 6m = 0$, lies in $(-2, 0)$.

Bound. cond^n



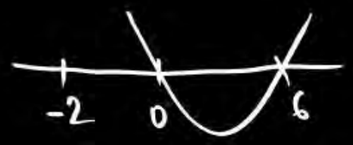
$x = 0$ is a root. ✓

$m^2 + 6m = 0$

$m = -6$ — Reject.

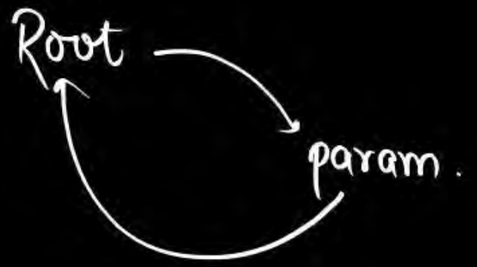
$x^2 - 6x = 0$

$x(x-6) = 0$ → 0, 6



$m = 0$ — Reject.

$x^2 = 0$ → 0, 0



Question [JEE Mains-2020]

The set of all real values of λ for which the quadratic equations, $(\lambda^2+1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0, 1)$ is

A $(-3, -1)$

B $(2, 4]$

C $(0, 2)$

D $(1, 3]$

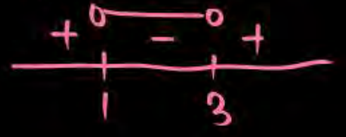


$\Rightarrow f(0) \cdot f(1) < 0$

$2 \cdot (\lambda^2+1-4\lambda+2) < 0$

$\lambda^2 - 4\lambda + 3 < 0$

$(\lambda-1)(\lambda-3) < 0$



$\lambda \in (1,3)$

fnf: $\lambda \in (1,3]$

Question [JEE Mains-2020]

The set of all real values of λ for which the quadratic equations, $(\lambda^2+1)x^2-4\lambda x+2=0$ always have exactly one root in the interval $(0,1)$ is

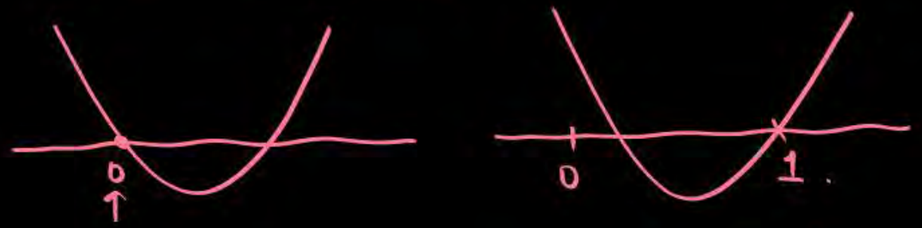
A $[1,3]$

B $[1,3)$

C $(1,3)$

D $(1,3]$ ✓

Boundary cond^n.



$x=0$

$2=0$

↓

n.p.

$x=1$ is root

$\lambda^2+1-4\lambda+2=0$

$\lambda^2-4\lambda+3=0$

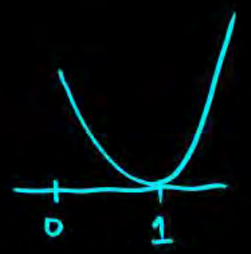
$\lambda=1,3$

✗ $\lambda=1$

$2x^2-4x+2=0$

$x^2-2x+1=0$

$(x-1)^2=0$ (1, 1)



$\lambda=3$

$10x^2-12x+2=0$

$5x^2-6x+1=0$

$(5x-1)(x-1)=0$ (1, $\frac{1}{5}$)



Question [JEE Mains-2020]

The set of all real values of λ for which the quadratic equations, $(\lambda^2+1)x^2 - 4\lambda x + 2 = 0$ always have exactly one root in the interval $(0, 1)$ is

A $[1,3]$

B $[1,3)$

C $(1,3)$

D $(1,3]$

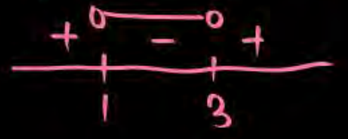


* $f(0) \cdot f(1) < 0$.

$2 \cdot (\lambda^2+1-4\lambda+2) < 0$

$\lambda^2 - 4\lambda + 3 < 0$

$(\lambda-1)(\lambda-3) < 0$



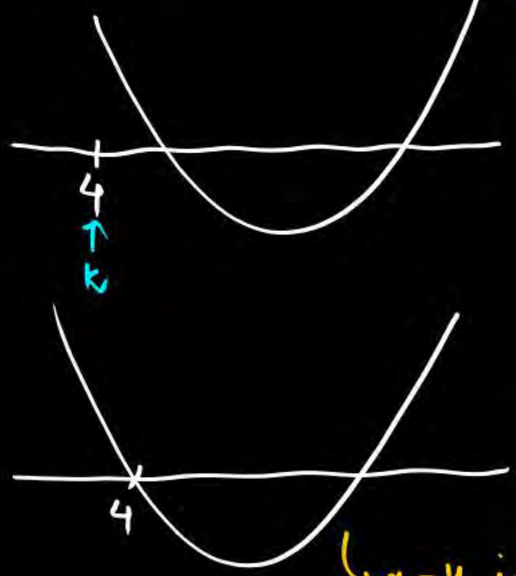
$\lambda \in (1,3)$

Question [IIT-JEE-2009]

The smallest value of k, for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

M-1

$D > 0$



① $D > 0$

② $-\frac{b}{2a} > 4$

③ $af(4) > 0$

$\} \cap$

↳ $x = 4$ is root $\longrightarrow 16 - 32k + 16(k^2 - k + 1) = 0$

$1 - 2k + k^2 - k + 1 = 0$

$k^2 - 3k + 2 = 0 \begin{cases} 1 \\ 2 \end{cases}$

$k = 1$ ✗

$x^2 - 8x + 16 = 0$

$(x-4)^2 = 0 \begin{cases} 4 \\ 4 \end{cases}$ ✗

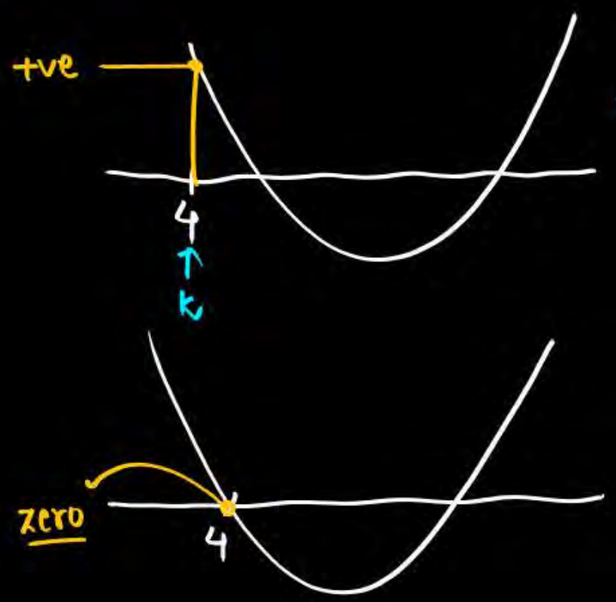
$k = 2$ ✓

$x^2 - 16x + 48 = 0 \begin{cases} 12 \\ 4 \end{cases}$ ✓

Question [IIT-JEE-2009]

The smallest value of k, for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

$D > 0$



M-2

① $D > 0$

② $-\frac{b}{2a} > 4$

③ $af(4) \geq 0$

$\cap$

• D is BY - (X)

Question [IIT-JEE-2009]

$k = 2$

The smallest value of k, for which both the roots of the equation, $x^2 - 8kx + 16(k^2 - k + 1) = 0$ are real, distinct and have values at least 4, is

4-3

$D > 0$

$$x = \frac{8k \pm \sqrt{64k^2 - 4 \times 16(k^2 - k + 1)}}{2 \times 1}$$

$$= \frac{8k \pm 8\sqrt{k-1}}{2} = 4\left(k \pm \sqrt{k-1}\right)$$

$4\left(k + \sqrt{k-1}\right)$ → larger root

$4\left(k - \sqrt{k-1}\right)$ → smaller root

def

$k - 1 > 0$

$k > 1$

$4\left(k - \sqrt{k-1}\right) \geq 4$

$k - \sqrt{k-1} \geq 1$

$k - 1 \geq \sqrt{k-1}$

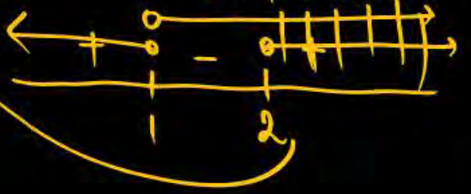
SBS.

$k^2 - 2k + 1 \geq k - 1$

$k^2 - 3k + 2 \geq 0$

$k \neq 1$

$k \in [2, \infty)$



Acidic Question

The values of k, for which the equation $x^2 + 2(k-1)x + k + 5 = 0$ posses atleast one positive root, are



A $[4, \infty)$

B $(-\infty, -1) \cup [4, \infty)$

C $[-1, 4]$

D $(-\infty, -1]$



THEORY OF HIGHER DEGREE EQUATIONS

$\equiv$ Sum

CUBIC EQUATION: (degree = 3 = no. of Root)

$$\underset{+}{ax^3} \underset{-}{+ bx^2} \underset{+}{+ cx} \underset{-}{+ d} = 0, \quad (a \neq 0)$$

Roots = α, β, γ.

S_1

Sum of roots taken one at a time $= \alpha + \beta + \gamma = \sum\alpha = \frac{-b}{a}$

S_2

Sum of product of roots taken two at a time $= \alpha\beta + \beta\gamma + \gamma\alpha = \sum\alpha\beta = \frac{+c}{a}$

Product of roots $= \alpha\beta\gamma = \frac{-d}{a}$

S_3

RAPID FIRE ROUND

(i) For : $7x^3 - 5x^2 + 11x + 19 = 0$

(a) $S_1 = \alpha + \beta + \gamma =$ $\frac{-(-5)}{7}$ $= 5/7$

(b) $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha =$ $\frac{+11}{7}$ $= 11/7$

(c) $S_3 = \alpha\beta\gamma =$ $\frac{-(19)}{7}$ $= -19/7$

(ii) For : $-2x^3 + 5x^2 + 8 = 0$

(a) $S_1 = \alpha + \beta + \gamma =$ $5/2$

(b) $S_2 = \alpha\beta + \beta\gamma + \gamma\alpha =$ 0

(c) $S_3 = \alpha\beta\gamma =$ 4

$-2x^3 + 5x^2 + 0x + 8 = 0$

$S_1 = \frac{-5}{-2} = 5/2$

$S_2 = \frac{0}{-2} = 0$

$S_3 = \frac{-8}{-2} = 4$

IMPORTANT NOTE FOR CUBIC

(i) Every odd degree polynomial equation with real coefficients always have 1 real root. (Imaginary roots exists in Pair)

(ii) Newton's formula : $S_n = p\,\alpha^n \pm q\,\beta^n \pm r\,\gamma^n$

Then : $aS_n + bS_{n-1} + cS_{n-2} + dS_{n-3} = 0$

$ax^3 + bx^2 + cx + d = 0$

$aS_n + bS_{n-1} + cS_{n-2} + dS_{n-3} = 0$

(iii) Factorisation of cubic: $ax^3 + bx^2 + cx + d = a(x-\alpha)(x-\beta)(x-\gamma)$ (identity)

(iv) Form a cubic whose root are α, β & γ.

$x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta+\beta\gamma+\gamma\alpha)x - (\alpha\beta\gamma) = 0$ ← α, β, γ

RAPID FIRE ROUND

(iii) Form a cubic whose roots are $-1, 3, 5$.

(iv) Form a cubic with rational coefficient and whose roots are: 5 & $3-\sqrt{2}$.

$x^3-(7)x^2+(7)x-(-15)=0$

$\alpha+\beta+\gamma=7$

$\alpha\beta+\beta\gamma+\gamma\alpha=-3+15-5=7$

$\alpha\beta\gamma=-15$

$x^3-7x^2+7x+15=0$ (-1, 3, 5)

$5,\ 3-\sqrt{2},\ 3+\sqrt{2}$

$\alpha+\beta+\gamma=11$

$\alpha\beta+\beta\gamma+\gamma\alpha=37$

$\alpha\beta\gamma=35$

$x^3-(11)x^2+(37)x-(35)=0$

Challenger Question

For what real values of a do the roots $x^2 - 2x - a^2 + 1 = 0$ lie between the roots $x^2 - 2(a+1)x + a(a-1) = 0$.



Challenger Question

Find the values of m for which the expression: $\frac{2x^2 - 5x + 3}{4x - m}$ can take all real values for $x \in R - \left\{\frac{m}{4}\right\}$

[Ans. : $m \in (4, 6)$]



DO IT BY YOURSELF (DIBY)

DIBY-61

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

[JEE MAINS - 2020]

(A) 18 (B) 36 (C) 9 (D) 27

[Ans. : (A)]

DIBY-62

If one root is square of the other root of equation $x^2 + px + q = 0$, then the relation between p and q is

[IIT-JEE-2004]

(A) $p^3 - q(3p - 1) + q^2 = 0$

(B) $p^3 - q(3p + 1) + q^2 = 0$

(C) $p^3 + q(3p - 1) + q^2 = 0$

(D) $p^3 + q(3p + 1) + q^2 = 0$

[Ans. : (A)]

LOCATION OF ROOTS

DIBY-63

Let $x^2-(m-3)x+m=0(m\in R)$ be a quadratic equation. Find the value of m for which

(i) both the roots are smaller than 2.

(ii) both the roots are greater than 2.

(iii) one root is smaller than 2 and the other root is greater than 2.

(iv) exactly one root lies in the interval (1, 2).

(v) both the roots lie in the interval (1, 2).

(vi) one root is greater than 2 and the other root is smaller than 1.

(vii) atleast one root lie in interval (1, 2).

(viii) atleast one root is greater than 2.

Ans. :

(i) $m\in(-\infty,1]$, (ii) $m\in[9,10)$, (iii) $m\in(0,\infty)$, (iv) $m\in(10,\infty)$,
(v) $m\in\phi$, (vi) $m\in\phi$, (vii) $m\in(10,\infty)$, (viii) $m\in[9,\infty)-\{10\}$

LOCATION OF ROOTS

DIBY-64

Find all values of a for which the equation $4x^2 - 2x + a = 0$ has two roots lie in the interval $(-1, 1)$.

[Ans. : $a \in \left(-2, \frac{1}{4}\right]$]

DIBY-65

The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval $(1, 2)$ and its other root lies in the interval $(2, 3)$, is [JEE Mains-2023]

(A) 2 (B) 0 (C) 1 (D) 3

[Ans. (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: (3 sittings of 1 HR each)

Exercise-02(Prabal): 1 to 10,13,16,18 to 25.

Exercise-03(Setu): 1 to 30 (except-13,16,17,21,25,26,29)

It is never about end result

It's always about Journey

Thank You

#futureIITians